



Course title: Project Management with Git			Credits	01
Course Code	25CSAE370	Course type	AEC	
Teaching Hours/Week(L:T:P:S)	0:0:2:0	Total Hours of Pedagogy in a semester	30 (00:00:28:02)	
CIE type	Lab CIE + Lab test	CIE Marks	50	
SEE Examination type	Practical	SEE Marks	50	
SEE Exam Hours	3	Total Marks	100	
Course Outcomes: At the end of the course the student will be able to: CO1. Execute basic Git repository commands to perform version control operations. CO2. Implement branch management strategies for collaborative software development. CO3. Make use of collaboration techniques using remote repositories and distributed version control systems. CO4. Apply Git tags, releases, and advanced Git operations for effective project management. CO5. Apply Git history modification techniques to maintain repository integrity and support project requirements.				
	Set of Experiments		Number of hours: 0:0:28:0	
1.	Setting Up and Basic Commands: Initialize a new Git repository in a directory. Create a new file and add it to the staging area, and commit the changes with an appropriate commit message.			
2.	Creating and Managing Branches: Create a new branch named "feature-branch." Switch to the "master" branch. Merge the "feature-branch" into "master."			
3.	Creating and Managing Branches: Write the commands to stash your changes, switch branches, and then apply the stashed changes.			
4.	Collaboration and Remote Repositories: Clone a remote Git repository to your local machine.			
5.	Collaboration and Remote Repositories: Fetch the latest changes from a remote repository and rebase your local branch onto the updated remote branch.			
6.	Collaboration and Remote Repositories: Write the command to merge "feature-branch" into "master" while providing a custom commit message for the merge.			
7.	Git Tags and Releases: Write the command to create a lightweight Git tag named “v1.0” for a commit in your local repository.			
8.	Advanced Git Operations: Write the command to cherry-pick a range of commits from "source-branch" to the current branch.			
9.	Analyzing and Changing Git History: Given a commit ID, how would you use Git to view the details of that specific commit,			



	including the author, date, and commit message?
10.	Analyzing and Changing Git History: Write the Git command to display all commits made by the author "JohnDoe" between "January 1, 2025" and "December 31, 2025."
11.	Analyzing and Changing Git History: Write the command to display the last five commits in the repository's history.
12.	Analyzing and Changing Git History: Write the command to undo the changes introduced by the commit with the ID "abc123".
Suggested Learning Resources	
Text Books	
1. Version Control with Git, 3rd Edition, by Prem Kumar Ponuthurai, Jon Loeliger Released October 2022, Publisher(s): O'Reilly Media, Inc.	
Learning Resources	
2. Pro Git book, written by Scott Chacon and Ben Straub and published by Apress, https://git-scm.com/book/en/v2	
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0130944433473699842782_shared/overview	
4. Git & GitHub Crash Course for Beginners [2026] https://www.youtube.com/watch?v=mAFoROnOfHs	
5. Learn GitHub within GitHub https://learn.github.com/skills	
6. Complete GitHub Actions Course - From BEGINNER to PRO https://www.youtube.com/watch?v=Xwpi0ITkL3U	
7. GIT CHEAT SHEET https://education.github.com/git-cheat-sheet-education.pdf	
8. Learn Git and GitHub https://roadmap.sh/git-github	
9. Learn Git Branching learngitbranching.js.org	
Teaching-Learning Process (Innovative Delivery Methods):	
1. Demonstration-based learning.	
2. Hands-on programming and debugging sessions	
3. Problem-based learning through project activities	
4. Self-learning using online tutorials and digital resources	

**Assessment Details**

Assessment Component	Subcomponent	Maximum Marks	Weightage / Scaling	Evaluation Criteria / Remarks
Overall Assessment	Continuous Internal Evaluation (CIE)	50	50% of Total	Minimum passing marks: 20/50 (40%)
	Semester End Examination (SEE)	50	50% of Total	Minimum passing marks: 18/50 (35%)
	Overall Pass Requirement	100	100%	Students must secure 40/100 (40%) in CIE + SEE combined to earn credits.
CIE	Record / Journal	96	Scaled to 24 Marks (80%)	The record must contain all prescribed experiments. Each experiment is evaluated for 8 marks based on observation sheet, write-up, correctness, neatness, and timely submission using faculty-defined rubrics.
	Self-Learning	24	Scaled to 6 Marks (20%)	Students are required to study the prescribed YouTube playlist and complete a quiz consisting of 24 questions. Each question carries 01 mark.
	Laboratory Test	100	Scaled to 20 Marks (40%)	Conducted after completion of experiments. Evaluation includes test write-up, experiment conduction, procedure, and result (60%) and viva-voce (40%).
	Total CIE	-	50 Marks	Sum of scaled Record/Journal (24), Self-Learning (6), and Laboratory Test (20).



SEE	Practical Examination	100	Scaled to 50 Marks	Conducted jointly by two internal examiners appointed by the Head of the Institution. Minimum duration: 2 hours. The student selects one experiment from a jointly prepared question lot covering all syllabus experiments.
	SEE Evaluation Rubrics	100	Scaled to 50 Marks	Write-up – 20%, Conduction Procedure & Result – 60%, Viva-Voce – 20% (or course-specific rubrics jointly decided by examiners).
	Change of Experiment	-	Penalty Applicable	Change of experiment is permitted only once. If changed, 15% of the marks allotted to the procedure component shall be awarded zero.
Result	Credit Award	100	-	A student earns the course credits only after satisfying the minimum passing requirements in CIE (20/50), SEE (18/50), and Overall (40/100).

Self-learning (SL) components

Self-Learning (includes MOOCs, spoken tutorials, online educational resources etc.)

Module No.	Topic	Self-Learning Activity	Resource / Link	No. of Hours	Assessment Method
1	Quiz on GitHub	Self-study	https://www.youtube.com/watch?v=mAFoROnOfHs	2	Quiz (24 Marks)

**Rubrics for CIE – Continuous Internal Evaluation**

Components / CO Mapping	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Unacceptable (1)
Git Fundamentals and Repository Operations [CO1] [PO1, PO5]	Demonstrates excellent understanding and independently performs repository initialization, staging, committing and basic Git operations correctly.	Demonstrates good understanding and performs basic Git operations with minor errors.	Demonstrates basic understanding and performs operations with some errors or guidance.	Demonstrates limited understanding and requires significant assistance.	Unable to perform or explain basic Git operations.
Branching and Workflow Management [CO2] [PO2, PO3]	Independently creates, switches, manages and merges branches correctly.	Performs branch operations correctly with minor errors.	Performs basic branch operations with some guidance.	Requires significant assistance to perform branch operations.	Unable to perform basic branch operations.
Remote Collaboration and Repository Management [CO3] [PO2, PO5, PO9]	Efficiently performs clone, fetch, push, pull, rebase and remote repository operations correctly.	Performs remote operations correctly with minor errors.	Performs basic remote operations with some guidance.	Shows limited ability in remote repository operations.	Unable to perform or explain remote repository operations.
Tags, Releases and Advanced Git Operations [CO4] [PO4, PO11]	Correctly performs tagging, release-related and advanced operations such as	Performs advanced operations correctly with minor errors.	Performs basic advanced operations with some guidance.	Requires significant assistance to perform advanced operations.	Unable to perform or explain advanced Git operations.



	cherry-pick with clear understanding.				
Git History Analysis and Modification [CO5] [PO2, PO4, PO5, PO11]	Accurately analyzes repository history and correctly performs required history-related operations.	Performs history analysis and modification with minor errors.	Performs basic history operations with some guidance.	Shows limited ability to analyze or modify Git history.	Unable to analyze or perform Git history operations.
Experiment Demonstration and Documentation [CO1-CO5] [PO10, PO11]	Demonstrates experiments confidently and maintains a complete, accurate and well-organized record.	Demonstrates most experiments correctly and maintains a well-organized record with minor errors.	Demonstrates basic experiments and maintains an adequate record with some missing details.	Demonstrates limited understanding and maintains an incomplete or poorly organized record.	Unable to demonstrate experiments or maintain a proper record.

**RUBRICS for SEE – PRACTICAL EXAMINATION**

Components / CO Mapping	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Unacceptable (1)
Experiment Understanding and Planning [CO1-CO5]	Clearly understands the assigned experiment, selects appropriate commands and plans the solution independently.	Understands the experiment and plans the solution with minor guidance.	Demonstrates basic understanding but requires some guidance.	Shows limited understanding and requires significant guidance.	Unable to understand or plan the assigned experiment.
Implementation / Conduction [CO1-CO5]	Executes the experiment correctly and independently using appropriate Git commands and workflow.	Executes the experiment correctly with minor errors.	Executes major steps with some errors and guidance.	Requires significant assistance to complete the experiment.	Unable to execute the experiment.
Procedure and Result [CO1-CO5]	Follows the correct procedure and obtains, verifies and clearly interprets the expected result.	Follows the procedure and obtains the expected result with minor errors.	Follows the basic procedure but requires guidance to obtain or verify the result.	Demonstrates limited ability to follow the procedure or verify the result.	Unable to complete the procedure or obtain a meaningful result.
Write-up / Documentation [CO1-CO5]	Provides complete, accurate and well-organized write-up with	Provides a clear and mostly complete write-up with minor	Provides an adequate write-up with some missing details.	Write-up is incomplete or contains significant errors.	Unable to provide an appropriate write-up.



	appropriate commands, steps and results.	errors.			
Viva-Voce / Conceptual Understanding [CO1-CO5]	Clearly explains Git concepts, commands, workflow and results and answers questions confidently.	Explains concepts and commands correctly with minor gaps.	Demonstrate s basic conceptual understanding with some guidance.	Demonstrate s limited understanding and requires frequent prompting.	Unable to explain the concepts, commands or results.